ENTERPRISE BI & DATA SOLUTIONS SQUAD

COGNOS & SPRING BOOT

Crystal Reports Migration • IBM Cognos Analytics • Complex SQL Optimization • Java Spring Boot Embedding

Engagement: Crystal Reports to Cognos BI Modernization
Focus Location: Richardson, Texas Project Mandate
Architecture: Java Spring Boot Embedded Analytics & RDBMS
Model: 3-Specialist Solutions Squad (Architect + Database DBA + BI PM)

Cognos Analytics Modernization & Java Spring Boot Embedded Reporting

A senior enterprise BI and backend engineering squad dedicated to modernizing legacy Crystal Reports into high-performance, interactive IBM Cognos Analytics dashboards. Combining deep relational SQL performance tuning, Framework Manager / Data Module metadata modeling, and seamless RESTful integration into Java Spring Boot enterprise applications.

- BI Core: **IBM Cognos Analytics 11.x / 12.x • Framework Manager**
- Legacy Conversion: **Crystal Reports → Cognos Interactive Reports**
- Backend: **Java 21 / Spring Boot 3.x • RESTful APIs • Security Filters**
- Database: **Complex SQL • Index Optimization • Query Plan Tuning**
- Governance: **Multi-Tenant Row-Level Security • Zero Discrepancy**

1. EXECUTIVE SUMMARY & DELIVERY APPROACH

Bridging Legacy Reporting with Modern Cloud-Native Application Architecture

The Challenge of Crystal Reports Migration

COMPLEXITY

Migrating legacy Crystal Reports to IBM Cognos Analytics in an enterprise environment involves far more than simply recreating visual report layouts:

- **Embedded Business Logic:** Formulas, conditional suppressions, and subreports trapped in Crystal files must be deconstructed and translated into centralized semantic models.
- **SQL Query Bottlenecks:** Unoptimized legacy queries causing slow page loads, lock contention, and database strain under concurrent enterprise reporting loads.
- **Spring Boot Application Integration:** Seamlessly embedding Cognos dashboards into existing Java Spring Boot portals with unified OAuth2/JWT security and responsive UI containers.

The Triad Squad Advantage for Richardson

EXECUTION SQUAD

Rather than relying on a standalone report writer who lacks backend Java Spring Boot experience, our squad delivers end-to-end integration:

- **Lead Java Spring Boot & Integration Architect (Viranga Lakshan):** Connects Cognos RESTful services, Mashup Services, and embedded iframe/token auth into Spring Boot microservices.
- **Senior Database & SQL Performance Engineer (Janith Kavishka):** Deconstructs legacy queries, rewrites nested joins into optimized views, and optimizes execution plans.
- **BI Delivery Project Manager (Lithumitha Ravikumar):** Led enterprise multi-dashboard platforms (*Castle Visualizer* and *Guardian Funds*), managing QA reconciliation and stakeholder signoffs.

2. TECHNICAL CAPABILITY ALIGNMENT MATRIX

Direct Mapping to Richardson Texas Cognos BI Mandate

PROJECT MANDATE REQUIREMENT	SQUAD ENGINEERING STANDARD & APPROACH	VERIFIED PRODUCTION PRECEDENTS
Crystal Reports Modernization	Reverse-engineering formulas, parameters, and grouping in Crystal; recreating as interactive, prompt-driven IBM Cognos reports and drill-through dashboards.	Enterprise analytics consolidation across legacy databases on *Castle Visualizer* (Australia) and ERP systems.
Cognos Metadata & Data Modeling	Designing governed semantic layers using Cognos Framework Manager and modern Data Modules; establishing star/snowflake schemas, determinants, and query subjects.	Multi-tenant schema isolation, domain-driven data models, and enterprise database architecture.
Advanced SQL Tuning & RDBMS	Query execution plan analysis, index optimization, eliminating Cartesian joins, partitioning large tables, and implementing materialized reporting views.	High-volume transaction tracking and zero-downtime database optimization for SINGER Sri Lanka PLC (Amazon RDS).
Java Spring Boot Integration	Integrating Cognos Mashup Service (CMS) and REST APIs with Java Spring Boot; securing embedded reports via JWT token delegation and Spring Security filters.	Architected enterprise Java 21 / Spring Boot 3.4.5 microservices, Keycloak OAuth2/JWT security, and SAP ERP integration for DIMO Lanka.
Data Discrepancy & Performance QA	Automated pixel-and-data comparison scripts validating historical Crystal output against newly generated Cognos reports with 100% data reconciliation.	3-way automated financial reconciliation engine (*InvoiceIQ*) and QA acceptance validation for *Guardian Funds*.

Full US Central Time (CT) Collaboration & High-Velocity Delivery

MIGRATION METHODOLOGY & ARCHITECTURE

Structured Crystal-to-Cognos Modernization and Java Spring Boot Embedding

INTEGRATION

Tooling: IBM Cognos Analytics • Framework Manager
Backend: Java Spring Boot 3.x • REST APIs
Security: OAuth2 / JWT • Row-Level Data Filtering

3. CRYSTAL REPORTS TO COGNOS MODERNIZATION FRAMEWORK

A Disciplined, Error-Free 4-Stage Conversion Protocol

Stage 1: Catalog Audit & Logic Extraction

ANALYSIS

We inventory all existing Crystal Reports (`.rpt` files), cataloging data sources, parameters, record selection formulas, sub-reports, conditional suppressions, and visual formatting rules. Redundant and obsolete reports are identified to streamline the reporting catalog.

Stage 2: Semantic Modeling & SQL Tuning

DATA MODELING

Rather than copying raw SQL into Cognos, we build governed semantic models in **Framework Manager** or Cognos **Data Modules**. Multi-table joins are normalized, calculations are moved to the database layer, and query subjects are tuned for reusable enterprise reporting.

Stage 3: Interactive Cognos Report Authoring

MODERNIZATION

Reports are rebuilt as modern, responsive IBM Cognos reports: replacing static tabular printouts with interactive drill-throughs, dynamic prompts, conditional formatting, and executive summary KPI dashboards optimized for web and mobile consumption.

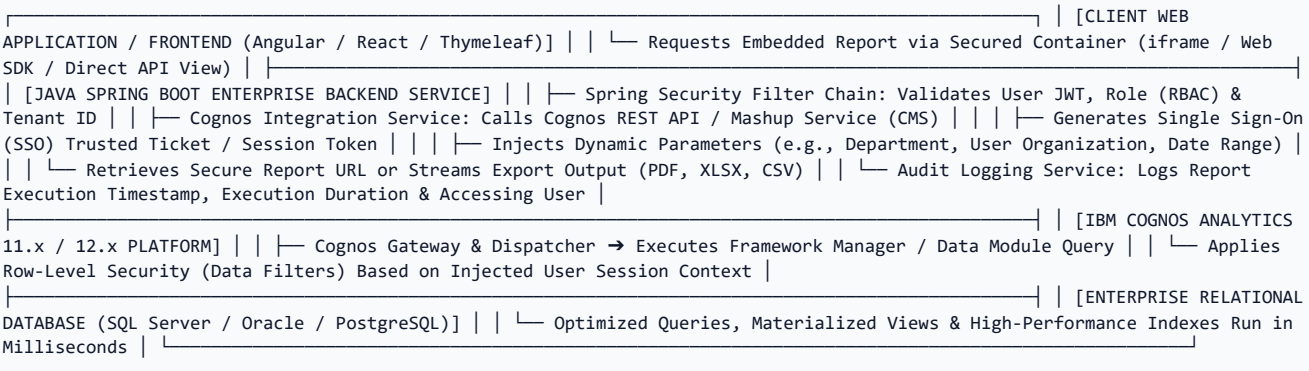
Stage 4: Data Validation & Discrepancy Audits

100% ACCURACY

We run side-by-side automated comparison tests matching historical Crystal Reports outputs against Cognos reports across identical date ranges and parameters, guaranteeing zero arithmetic drift, rounding errors, or missing edge-case rows.

4. JAVA SPRING BOOT EMBEDDED ANALYTICS ARCHITECTURE

Seamlessly Integrating Cognos Reporting into Your Enterprise Backend



5. ADVANCED SQL PERFORMANCE & QUERY OPTIMIZATION

Slashing Execution Times on Enterprise Reporting Databases

Execution Plan Analysis

QUERY TUNING

Diagnosing slow SQL by inspecting execution plans: eliminating full table scans, resolving hash match bottlenecks, and rewriting inefficient correlated subqueries.

Indexed & Materialized Views

OPTIMIZATION

Designing targeted composite indexes and materialized reporting views that pre-aggregate massive transaction datasets, delivering instant report responses.

Row-Level Security (RLS)

DATA ISOLATION

Enforcing tenant and departmental security directly at the data model level, guaranteeing users only view authorized enterprise data without duplicating report layouts.

ENTERPRISE ANALYTICS & GLOBAL PRECEDENTS

TRACK RECORD

Production BI Dashboards, Java Spring Boot Systems, and Database Performance

Domain Reach: Enterprise Analytics, ERP & FinTech
Backend Tech: Java 21, Spring Boot 3.x, REST APIs
Databases: SQL Server, Oracle, MySQL, Amazon RDS

6. VERIFIED ENTERPRISE PRODUCTION CASE STUDIES

Real International Systems Delivered by Squad Members

Castle Visualizer — Enterprise Data & Multi-Dashboard Analytics Platform

Client: Castle Group (Australia) • PM: Lithumitha Ravikumar

ENTERPRISE BI

Cross-System Enterprise Analytics Consolidation: Led the end-to-end delivery of an enterprise multi-dashboard analytics platform, consolidating disparate operational data from multiple legacy systems into a unified, high-performance reporting environment.

- **Stakeholder Requirements & Data Modeling:** Managed comprehensive requirements definition, translating complex corporate KPIs and operational metrics into standardized data structures.
- **Sprint Coordination & Quality Assurance:** Coordinated cross-functional engineering teams across data integration, database design, and frontend dashboards to deliver real-time visual insights for C-suite decision-makers.

DIMO Enterprise Gate Pass & Logistics Management System

Client: DIMO Lanka (Enterprise Logistics) • Lead: Viranga Lakshan

JAVA SPRING BOOT 3.X

Mission-Critical Java Microservices Architecture: Architected API Gateway and core domain microservices built with Java 21 and Spring Boot 3.4.5, managing complex approval hierarchies, audit event streaming, and enterprise ERP integration.

- **Spring Security & Keycloak OAuth2/JWT:** Implemented centralized token verification, role-based access control (RBAC), and secure REST API services connecting web and mobile clients.
- **Enterprise Integrations & Event Logging:** Integrated directly with SAP ERP, Microsoft Graph / Azure AD, and Spring Kafka publishing immutable audit logs for all transactions.
- **MySQL Database Migrations:** Managed relational schema design and automated database migrations using Flyway, containerizing services via Docker and CI/CD pipelines.

Financial Reporting & DB Optimization

SINGER Sri Lanka PLC • Janith Kavishka

SQL OPTIMIZATION

Engineered batch-wise financial transaction tracking and real-time reconciliation systems. Executed full data migration and continuous query tuning on Amazon RDS (MySQL) for nationwide enterprise retail operations.

Guardian Funds Analytics Dashboard

Client: Guardian Funds (Australia)

PORTFOLIO BI

Managed delivery of an investment analytics dashboard consolidating financial metrics for senior stakeholder decision-making. Enforced rigorous QA acceptance criteria validating reporting calculations.

PHASED ROADMAP & SQUAD PROFILES

EXECUTION

Structured Delivery Plan, Team Capabilities, and Seamless Richardson Collaboration

Timeline: 8-Week Phased Migration
Timezone: Overlap with US Central Time (CT)
Handover: Full Technical Documentation & Runbooks

7. 8-WEEK PHASED MODERNIZATION & INTEGRATION ROADMAP

Systematic Execution Guaranteeing Zero Interruption to Business Operations

Phase 1: Catalog Audit & SQL Mapping — WEEKS 1–2

- Inventory all legacy Crystal Reports (`.rpt`), identifying parameters, grouping, and sub-reports.
- Extract and document underlying SQL queries, stored procedures, and database dependencies.
- Establish baseline report runtime benchmarks and identify primary query bottlenecks.
- Deliver Crystal-to-Cognos Migration Matrix and prioritize high-value operational reports.

Phase 2: Data Modeling & Query Tuning — WEEKS 3–4

- Build governed metadata models using Cognos **Framework Manager** and Data Modules.
- Refactor complex SQL queries: optimize joins, create targeted indexes, and design views.
- Configure row-level security filters, user parameter prompts, and multi-tenant determinants.
- Conduct database query performance benchmarks demonstrating query speed improvements.

Phase 3: Cognos Authoring & Java Integration — WEEKS 5–6

- Author modern interactive Cognos reports, KPI summaries, and drill-through dashboards.
- Collaborate with backend developers to integrate Cognos RESTful / Mashup APIs into Java Spring Boot.
- Implement secure report embedding with Spring Security, passing dynamic JWT user tokens.
- Enable automated report scheduling and export delivery pipelines (PDF, Excel, CSV).

Phase 4: UAT, Reconciliation & Handover — WEEKS 7–8

- Execute automated data reconciliation tests comparing Crystal historical output vs. Cognos output.
- Partner with business analysts and product managers to conduct User Acceptance Testing (UAT).
- Resolve remaining edge-case discrepancies, performance bottlenecks, and UI adjustments.
- Deliver comprehensive technical documentation, data dictionary, and developer runbooks.

8. MEET THE BI & INTEGRATION ENGINEERING SQUAD

Senior Systems & Data Engineering Specialists Dedicated to Your Success

Viranga Lakshan

LEAD ENTERPRISE ARCHITECT & JAVA LEAD

Associate Software Engineer, Arimac • 4+ Years Enterprise Systems

Specializes in Java 21 / Spring Boot 3.x backend architecture, REST API design, OAuth2/JWT security, system integrations, and report embedding within enterprise portals.

Java Spring Boot | REST APIs | OAuth2 / JWT | API Gateways

Janith Kavishka

SENIOR DATABASE & SQL ENGINEER

Programmer, DMS Electronics • Enterprise Core Systems & Database Operations

Expert in relational database design, complex SQL tuning, query execution plan analysis, index optimization, data reconciliation, and high-volume reporting architecture.

Complex SQL Tuning | RDBMS Design | Execution Plans | Data Modeling

Lithumitha Ravikumar

BI PROJECT MANAGER & SYSTEMS ANALYST

Associate Project Manager, Arimac • Enterprise SaaS & Analytics Delivery

Leads requirements gathering, sprint coordination, cross-team collaboration, and QA acceptance criteria validation across enterprise analytics and reporting platforms.

BI Project Management | Requirements Analysis | QA Validation | Sprint Delivery

Next Steps & Technical Architecture Discussion

We welcome a technical discovery call with your engineering and reporting leadership to review sample Crystal Report definitions (`.rpt`), evaluate your database schema, and demonstrate our Java Spring Boot Cognos integration framework. We are ready to begin immediately upon engagement.